TITAN EYE EYEWEAR E-COMMERCE

DBMS TASK – SALES AND CUSTOMER ANALYTICS SYSTEM

Objectives

1. Apply COUNT(), SUM(), AVG(), MIN(), and MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis

Aggregate Functions

Definition

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.

1.1 COUNT()

Find the total number of orders.

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
           5

SQL>
```

Purpose: Counts the total number of orders in the Orders table.

1.2 SUM()

Find the total sales amount.

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
      24996
```

Purpose: Calculates the total sales generated by Titan Eye.

1.3 AVG()

Find the average order amount**.**

```
SQL> SELECT AVG(Total_Amount) AS Average_Order
  2  FROM Orders;

AVERAGE_ORDER
-------------
       4999.2
```

Purpose: Calculates the average amount spent per order.

1.4 MIN()

Find the lowest order amount.

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order
  2  FROM Orders;

MINIMUM_ORDER
-------------
         1999
```

Purpose: Displays the lowest order amount.

1.5 MAX()

Find the highest order amount.

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order
  2  FROM Orders;

MAXIMUM_ORDER
-------------
         7499
```

Purpose: Displays the highest order amount.

2. Generate Total Sales Reports

A sales report can be generated using the SUM() function on the Total_Amount column.

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
      24996
```

Purpose: This query calculates the total revenue generated from all Titan Eye orders.

2.1 Customer-wise Sales Report

A customer-wise sales report displays the total purchase amount of each customer.

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      SUM(O.Total_Amount) AS Total_Purchase
  5  FROM Customer C, Orders O
  6  WHERE C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.Customer_Name
  8  ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        101 Arun Kumar                                               4500
        102 Priya Sharma                                             4999
        103 Kavin Raj                                                5999
        104 Divya S                                                  1999
        105 Rahul Das                                                7499
```

Purpose: To calculate how much each customer has purchased.

3.Top Customers Based on Purchase Amount

This query identifies customers who have made the highest purchases.

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      SUM(O.Total_Amount) AS Total_Purchase
  5  FROM Customer C, Orders O
  6  WHERE C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.Customer_Name
  8  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        105 Rahul Das                                                7499
        103 Kavin Raj                                                5999
        102 Priya Sharma                                             4999
        101 Arun Kumar                                               4500
        104 Divya S                                                  1999
```

Purpose: To find the highest-spending customers.

4. Best-Selling Products

This query identifies the products sold in the highest quantity.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OI.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM Product P, Order_Item OI
  6  WHERE P.PRODUCT_ID = OI.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       101
Titan Classic Frame
                  2

       103
Titan Aviator Sunglasses
                  1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

       104
Titan Round Sunglasses
                  1
```

```
       104
Titan Round Sunglasses
                    1

       102
Titan Metal Frame

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
TOTAL_QUANTITY_SOLD
--------------------
                    1
```

Purpose: To find which products are selling the most.

4.1 Product Sales Amount

Product sales amount is calculated by multiplying the quantity sold by the product price and using the SUM() function

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OI.QUANTITY * OI.PRICE) AS TOTAL_SALES
  5  FROM Product P, Order_Item OI
  6  WHERE P.PRODUCT_ID = OI.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY TOTAL_SALES DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
TOTAL_SALES
-----------
       101
Titan Classic Frame
       9000

       103
Titan Aviator Sunglasses
       5999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
TOTAL_SALES
-----------
```

```
       102
Titan Metal Frame
       4800

       104
Titan Round Sunglasses

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
TOTAL_SALES
-----------
       1999
```

Purpose: To calculate the total sales amount generated by each product.

5.Category-wise Sales Analysis

Category-wise sales analysis uses JOIN and SUM() functions to calculate the total sales amount for each product category.

```
SQL> SELECT
  2      C.CATEGORY_ID,
  3      C.CATEGORY_NAME,
  4      SUM(OI.QUANTITY * OI.PRICE) AS TOTAL_SALES
  5  FROM Category C, Product P, Order_Item OI
  6  WHERE C.CATEGORY_ID = P.CATEGORY_ID
  7  AND P.PRODUCT_ID = OI.PRODUCT_ID
  8  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
  9  ORDER BY TOTAL_SALES DESC;

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
          1
Eyeglasses
      13800

          2
Sunglasses
       7998

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
```

Purpose: To identify which product category generates the highest sales revenue.